



Jevons’ Paradox and Artificial Intelligence

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Jevons’ Paradox

Jevons argued that as the steam engine made energy cheaper, coal-powered production would become more accessible across the economy. Greater efficiency would not help conserve coal but would rather spur further demand for it.

If cost savings are passed on to consumers, and demand is sufficiently elastic, Jevons’ paradox suggests that when technological progress increases the efficiency with which a resource is used, falling costs may stimulate demand so strongly that total consumption rises.

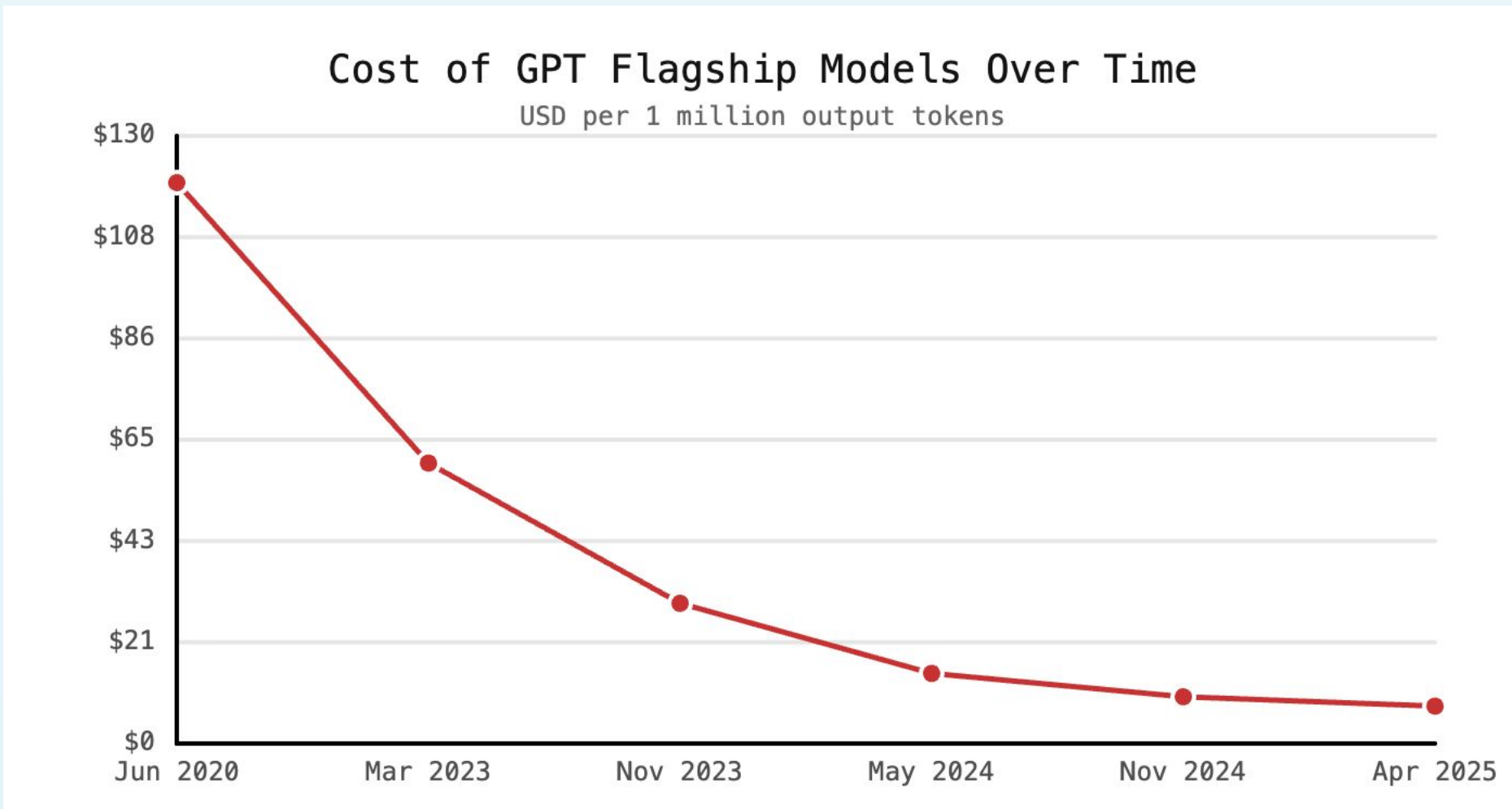
Applicability to Artificial Intelligence

Today, the resource is not coal but cognitive capital. Artificial intelligence has made cognitive work cheaper, just as the Watt steam engine did to energy.

However, Jevons’ coal question and the artificial intelligence question are not identical. While coal was a factor of production, it was also a complement to human labour, whereas artificial intelligence is a direct substitute. When cognitive work becomes cheaper, it threatens jobs.

Jevons’ paradox, therefore, cuts in two directions. On the one hand, the adoption of artificial intelligence may lead to increased demand for AI-assisted output, thereby protecting and creating jobs. On the other hand, adoption could also displace workers. Whether expansion or displacement wins depends on the elasticity of demand in each market segment.

Introducing Cost Savings



For OpenAI’s flagship model, output costs fell from \$60 per million output tokens in March 2023 to \$8 by April 2025. This is a 87% decrease in merely two years. Compared with GPT-3, that’s a 93% decrease.

In the UK, amongst many developed economies, white-collar jobs and services account for the majority of gross domestic product. This means the economy is particularly at risk of structural unemployment stemming from a wave of AI-driven job replacements.

Introducing Elasticities of Output

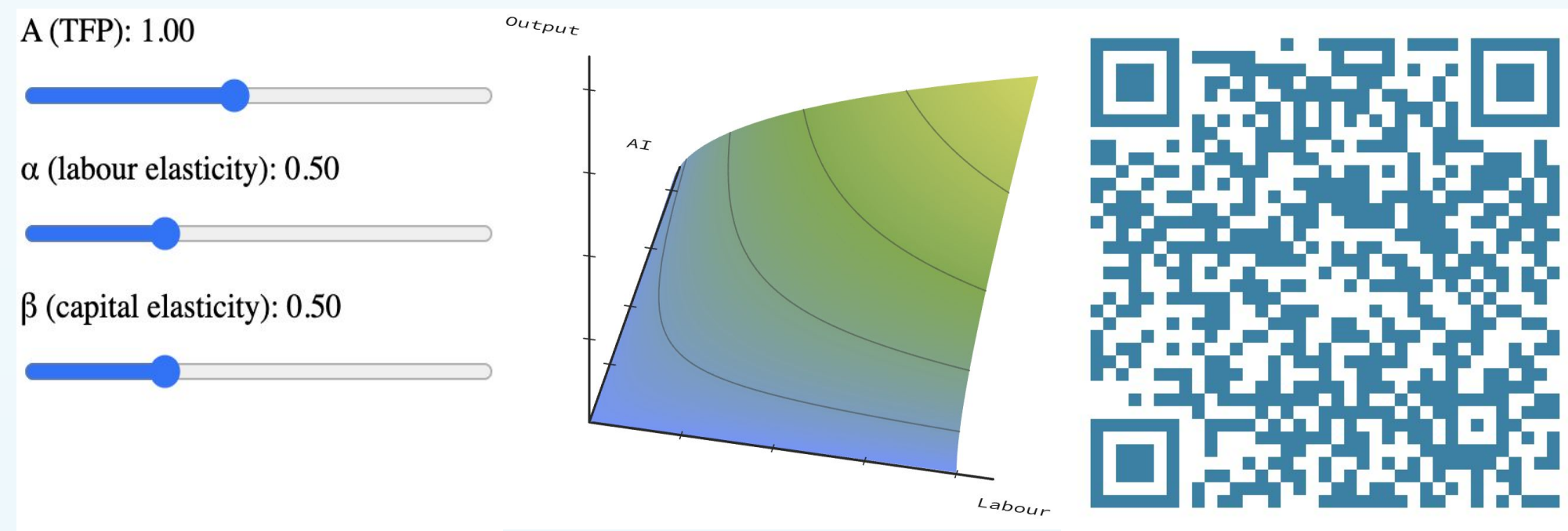
Given that demand for final goods is price-elastic, we would expect demand for intelligence to rise quickly as efficiency is improved.

One aspect is the substitutability of labour to agentic AI (p).

Y is output, L is labour input, K is AI input, A is multi-factor productivity, α is labour elasticity of output, and β is capital elasticity of output.

$$Y(L,K)=AL^{\alpha}K^{\beta}$$

We use Cobb-Douglas Functions to visualise the indifference curves of a firm integrating large language models alongside human labour.



Another aspect is the degree to which labour can reallocate to sectors where there is a greater need for humans than AIs (μ).

In market segments where workers are most susceptible to AI replacement, can they readily transition to ‘AI-proof’ employment, like manual jobs?

Some examples of jobs that are unlikely to be substituted by agentic AI include "cooks, motorcycle mechanics and lifeguards” (Anthropic, 2026).

Integrating with the WS-PS Model

Higher cost savings shift the PS curve upward: artificial intelligence increases productivity, reducing production costs. As a result, firms can profitably offer higher real wages at any level of employment.

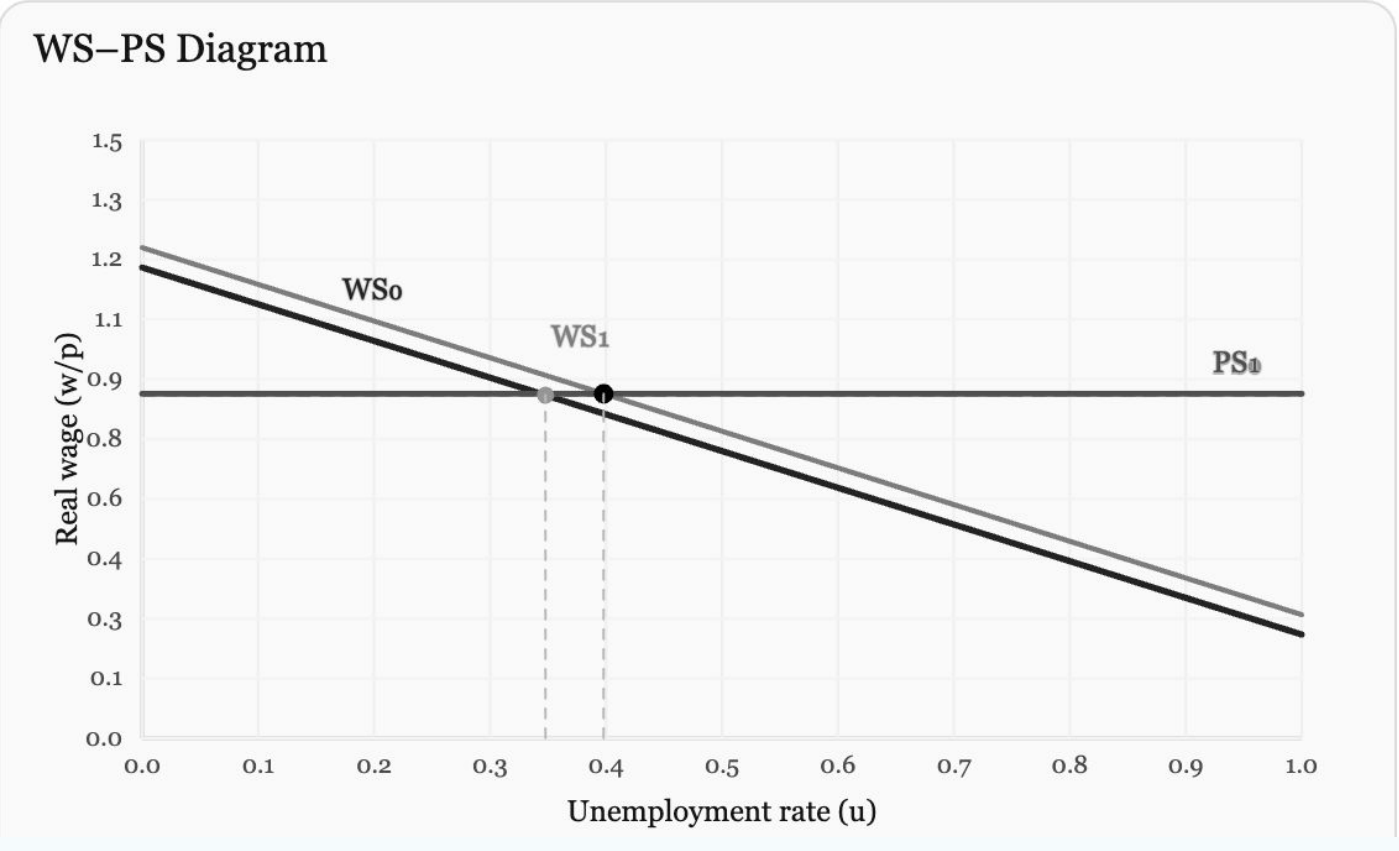
Higher pass-through amplifies the PS shift: when a higher portion of cost savings is passed on to consumers, this reduces the price markup, thereby raising the real wage firms can pay.

Higher demand elasticity shifts the PS curve upward: the price markup is inversely related to demand elasticity.

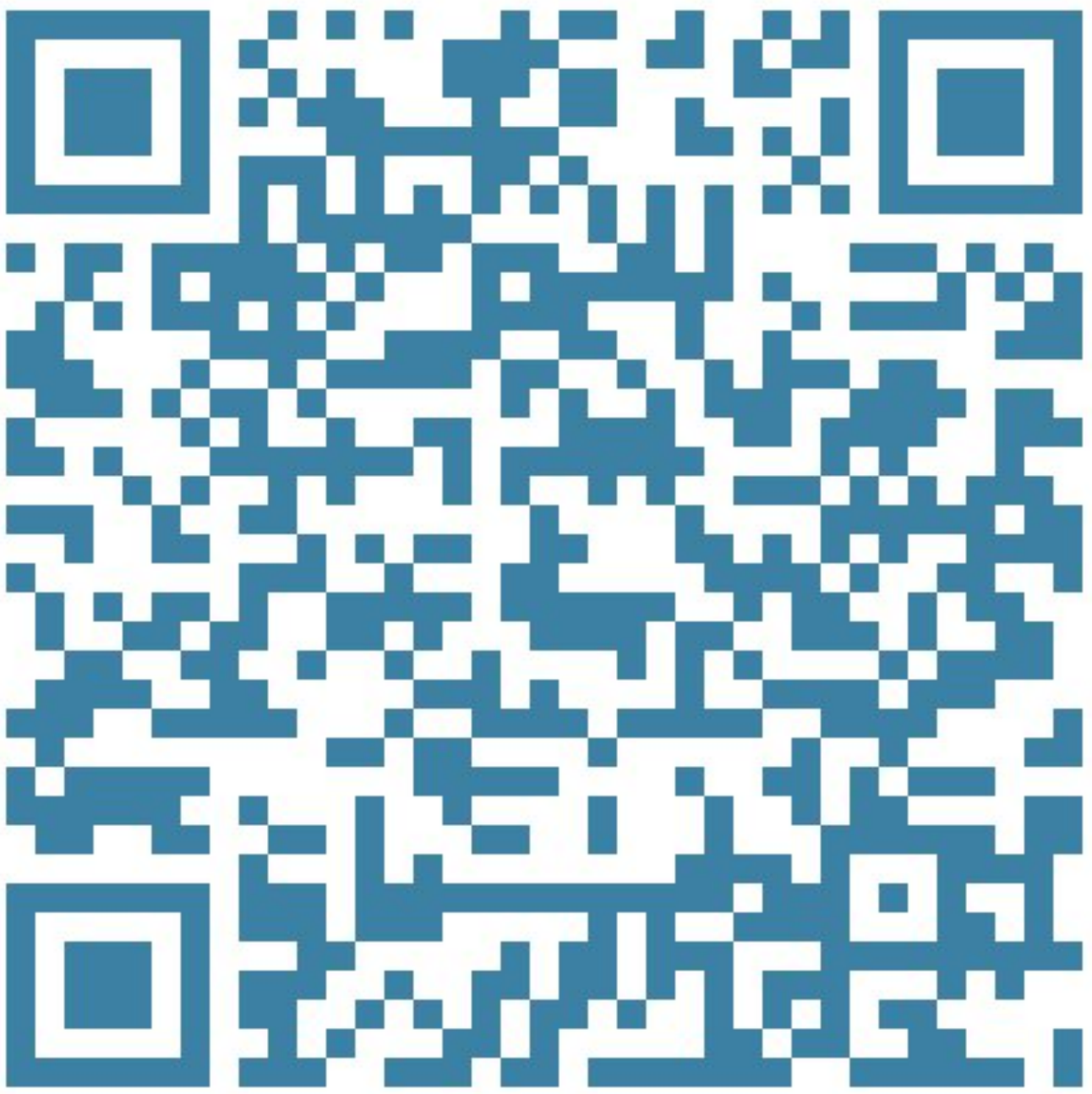
Higher replacement rate shifts the WS curve downwards: workers lose bargaining power as they become more easily replaced by artificial intelligence.

Higher mobility shifts the WS curve upward: it increases workers’ bargaining power.

Developing a Model



Play Me!



Our Variables

$s \in [0, 1]$, proportional unit cost saving from AI
 $\theta \in [0, 1]$, share of cost savings passed to consumers
 ϵ , price elasticity of demand $\epsilon < 0$
 ρ , replacement rate
 l , labour per unit
 L , employment
 μ , labour mobility

Our Modelling Assumptions

The model assumes an isoelastic demand function, such that $Q = kP^{\epsilon}$. This means demand responds proportionally to price changes according to the elasticity . So $Q_0 = kP_0^{\epsilon}$ and $Q_1 = kP_1^{\epsilon}$.

Applying Intuitions

AI reduces production costs by s , where firms pass a share θ of those savings to consumers.

$$P_1 = P_0(1 - \theta s)$$

AI replaces a share ρ of labour per unit.

$$l_1 = (1 - \rho)l_0$$

Lower prices increase demand

$$Q_1 = Q_0(1 - \theta s)^{\epsilon}$$

Total employment becomes

$$L_1 = L_0(1 - \rho)(1 - \theta s)^{\epsilon}$$

Let $G \equiv (1 - \rho)(1 - \theta s)^{\epsilon}$

$$\Delta L = L_1 - L_0 = L_0(G - 1)$$

G summarises the net employment effect of AI.

With mobility μ

$$\Delta L < 0 : \Delta U = -(1 - \mu)(L_0(G - 1))$$

$$\Delta L \geq 0 : \Delta U = -\mu(L_0(G - 1))$$

Therefore

$$\Delta L < 0 : \Delta U = -(1 - \mu)(L_0((1 - \rho)(1 - \theta s)^{\epsilon} - 1))$$

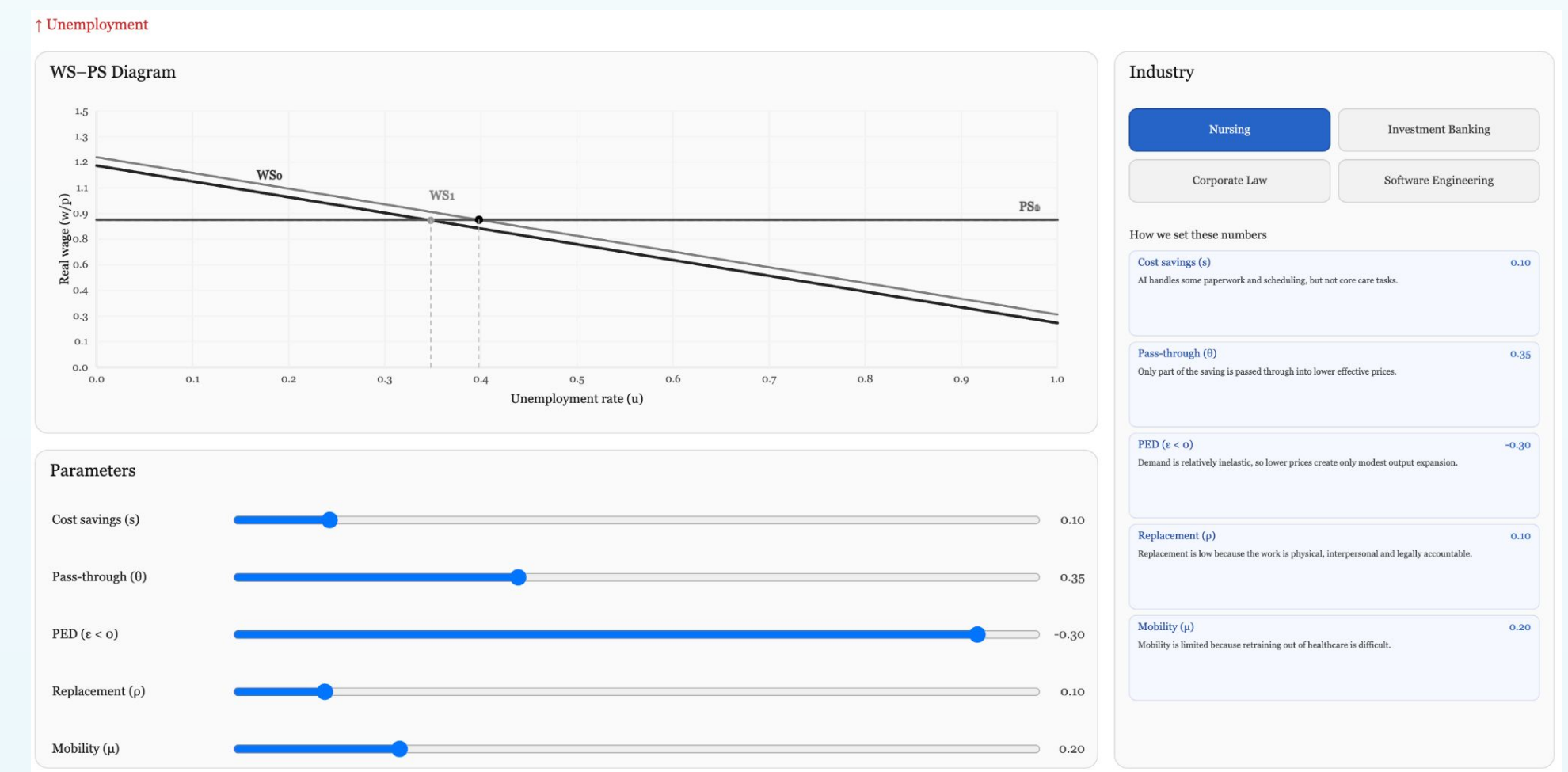
$$\Delta L \geq 0 : \Delta U = -\mu(L_0((1 - \rho)(1 - \theta s)^{\epsilon} - 1))$$

A Look at Nursing

In nursing, demand for healthcare is highly inelastic. Lower costs are unlikely to stimulate a sufficiently large increase in demand to offset potential job displacement.

That said, three important factors are not accounted for in our model:

1. In many countries, such as the UK, healthcare is publicly provided, so consumers are not faced with a clear price tag.
2. Healthcare systems, including those in the UK and the US, face a substantial backlog: there is persistent excess demand for nurses.
3. Ageing populations are expected to increase demand for healthcare services over time.



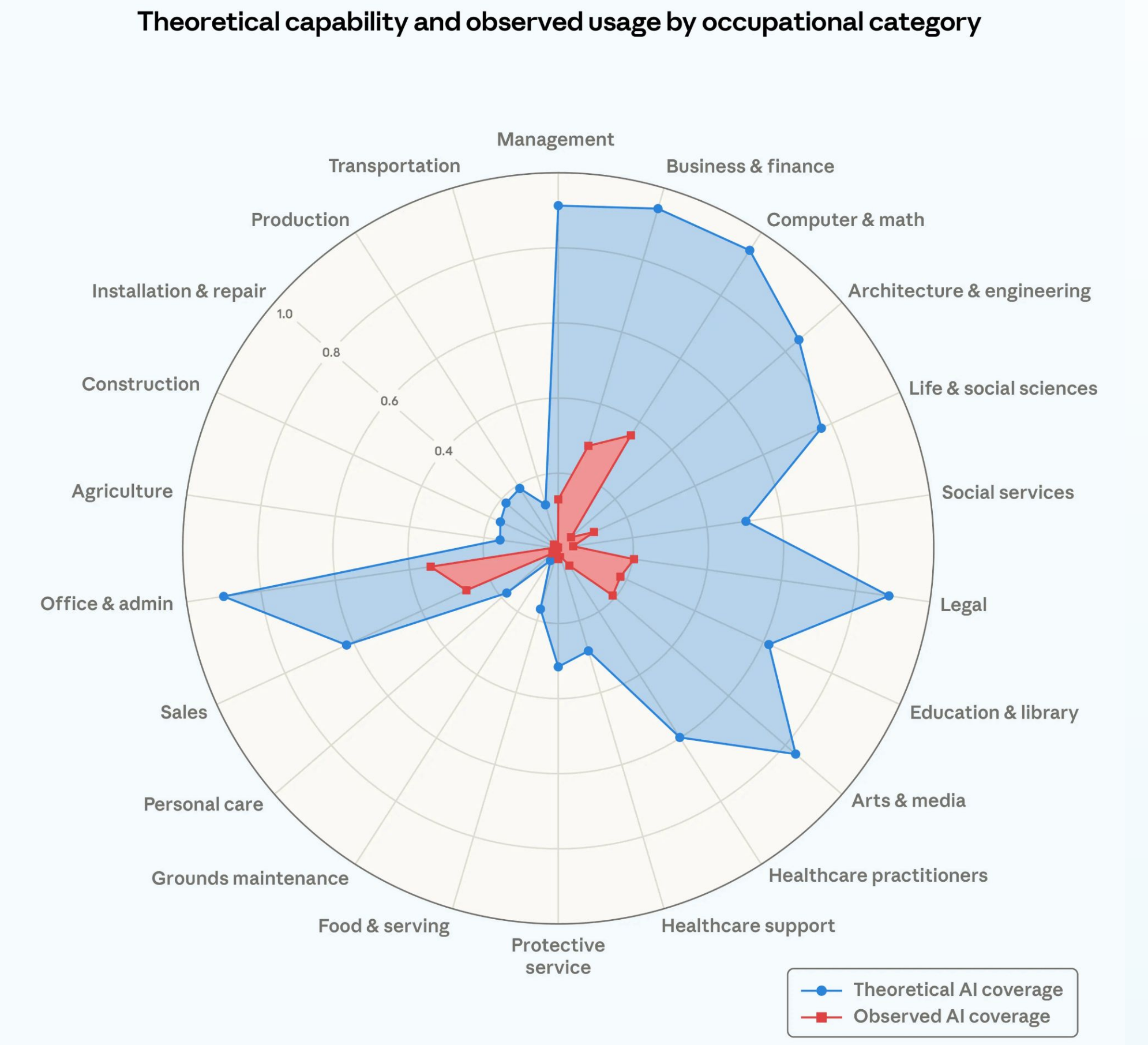
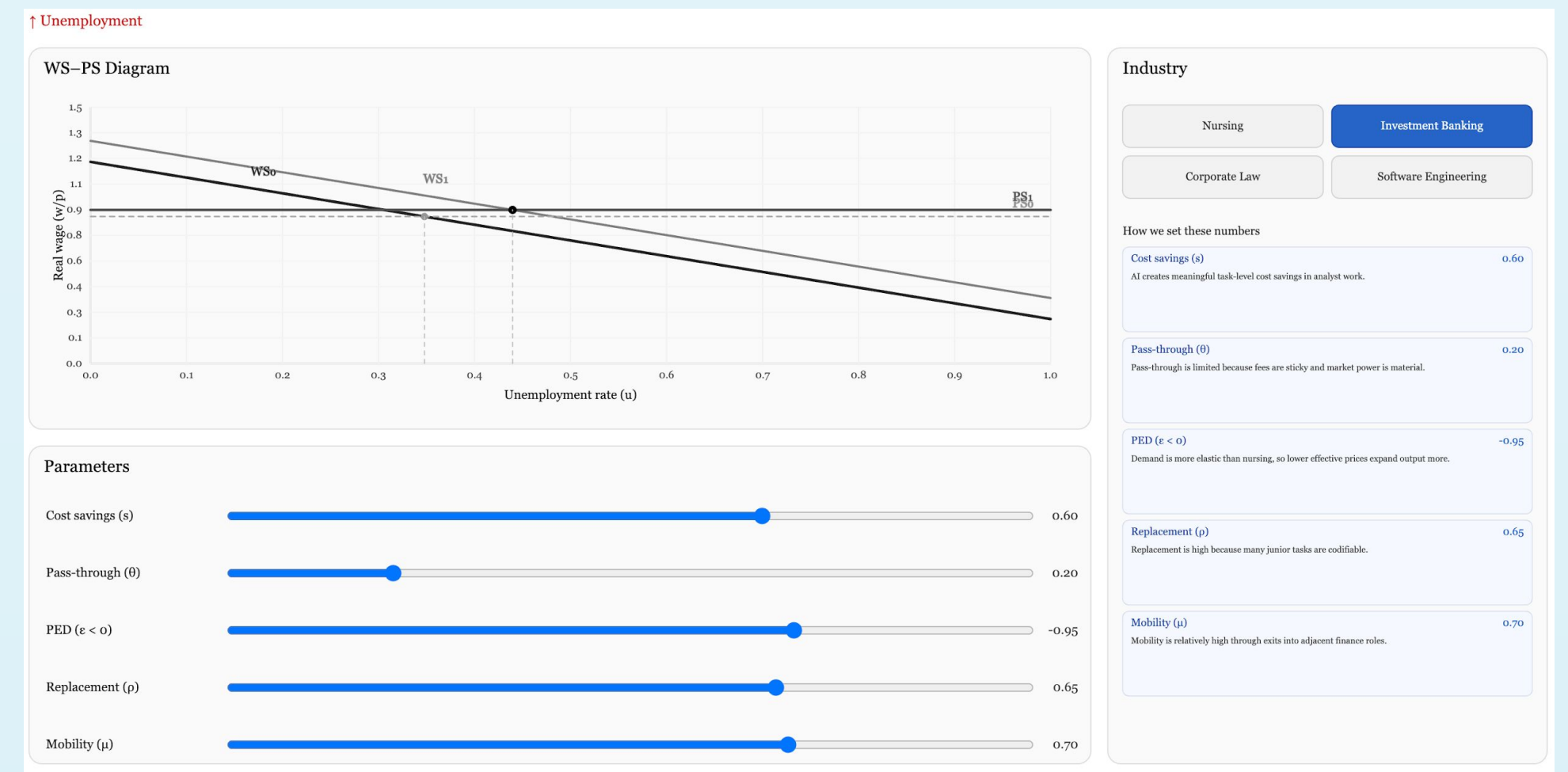
A Look at Investment Banking

Demand for investment banking services is expected to grow, and junior-level workflows are increasingly being automated. As a result, cost savings at the entry level are likely to be substantial.

High labour replacement rates mean that these efficiency gains are unlikely to translate into proportional increases in employment. Within the model, this implies a strong downward shift in labour demand for junior roles and a subsequent rise in unemployment.

However, a rise in unemployment may be mitigated by job creation in other sub sectors of the financial industry.

While artificial intelligence may reduce the number of entry-level jobs in the short run, it might not generate persistent unemployment - especially if the government regulates the use of artificial intelligence.



What This Means for Us

Jevons’ insight was that efficiency improvements do not conserve a resource but rather unleash greater demand for it. AI is doing exactly this, but to our ability to think, analyse, and create. Our model gives you 3 questions to ask about any sectors, including your own future career.

1. **Is demand elastic?**
2. **How substitutable are your tasks?**
3. **Can you move?**

Historically, the workers hit the hardest weren’t always the most substitutable, but the most substitutable with the lowest mobility. High replacement rates and low labour mobility - that combination is where structural employment is likely to emerge.

Even when employment holds up, the WS-PS model predicts a fall in real wages when the replacement rate is high. When the WS curve shifts down faster than the PS curve shifts up, the new Nash equilibrium delivers higher output, but a small labour share contributes to it.

However, this depends on labour protection laws. If a positive supply shock coincides with rising worker bargaining power, falling wages may be mitigated. The critical point is whether worker protections can close the bargaining gap. Without deliberate policy intervention, history suggests they rarely do.

References

1. [OpenAI API pricing announcements \(2020–2025\)](#)
2. [Anthropic Labor Market Impacts Report](#)
3. [CORE ECON The Economy 2.0](#)
4. [Wikipedia Article on Jevons Paradox](#)